

Sarthak Jain

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Published researcher who makes **AI and signal-processing workloads run fast on resource-limited hardware**: shipped a quantized neural net on a microcontroller in **under 8 ms**, built a pipelined **RISC-V CPU**, and aligned brain recordings to **under 2 ms**.

EDUCATION

Carnegie Mellon University / *B.S. Electrical & Computer Engineering and Robotics* / **GPA 3.8/4.0** Expected May 2028

- **Coursework:** Signals & Systems, Structure & Design of Digital Systems, Introduction to Computer Systems, Embedded Systems, Computer Vision, Fundamentals of Controls, Electronics & Circuits, Principles of Imperative Computation, Linear Algebra

EXPERIENCE

Scout Me / *Embedded ML Engineer (Remote)* 2024 – 2025

- Built a wearable vest that tracks an athlete's vital signs during play, fusing **7 sensor channels** (heart rate, SpO2, HRV, temperature, motion). **Now used by 500+ blind football players**.
- Movement was corrupting the readings; added a denoising autoencoder and adaptive **Kalman filter** that cut that noise **43%**.
- **Pruned and INT8-quantized** the model so it runs on the vest's microcontroller instead of a phone: **under 8 ms per reading** on a **sub-milliwatt** power budget.

AirLab, CMU Robotics Institute / *Robotics Autonomy Researcher* May 2026 – Present

- Taught a drone to sense a nearby wall's distance and angle from its **own motor and IMU data alone**. No camera, no LiDAR.
- Trained a **physics-informed PyTorch** model on **120,000 simulated flights** from an ANSYS CFD digital twin instead of crashed hardware; transferred it to the real aircraft at **under 3 cm** and **under 4°** error.

ForAll Lab, Carnegie Mellon / *Neuroscience / Signals Engineer* Summer 2026

- A brain-implant study recorded on several machines whose clocks drifted apart, so the data no longer lined up. Wrote the correction (least-squares drift model in **NumPy/SciPy**) that holds every stream **within 2 ms**, cutting timing spread **~30×** with **100% event markers recovered**.

Kantor Lab, Carnegie Mellon / *Research Assistant* 2025 – Present

- Deployed real-time **YOLOv8** inference on an autonomous farm robot that finds diseased apple branches and directs a robotic arm to mark them (**ROS 2**); team **won the Farm Robotics Challenge (\$25K)**.

Carnegie Mellon University, ECE / *Teaching Assistant, 18-290 Signals and Systems* Fall 2026

- Teach recitations and office hours for **100+ students**: Fourier and Laplace transforms, the FFT, sampling, and filter design. Write and grade problem sets and exams.

PROJECTS

Pipelined RISC-V Processor | *SystemVerilog, computer architecture, Vivado, QuestaSim* 2025

- Built a working CPU (RV32I) from the instruction set up, with hazard-detection and forwarding logic that keeps **five instructions in flight at once** correct. Synthesized in Vivado; verified cycle by cycle with **0 hazard failures**.

Shadow Fight | *C++17, QNX RTOS, OpenCV, dual Raspberry Pi, real-time systems* 2025

- A game you play against your own shadow. Two Raspberry Pis on the **QNX real-time OS** read your silhouette from a camera and drive an opponent that reacts in **under 80 ms**, linked by a lean **50-byte UDP protocol**.

AI Air Hockey | *Computer vision, real-time control, OpenCV, embedded C* 2025

- A robot that plays air hockey against a person: an overhead camera tracks the puck, predicts its path (including wall bounces), and drives the paddle to meet it in **under 50 ms**. Profiled the full camera-to-motor loop to remove bottlenecks.

PUBLICATION & LEADERSHIP

Photovoltaic Panel Efficiency / *Co-Author, international renewable-energy journal* 2023 – 2024

- Modeled solar irradiance across spectral (wavelength) and angular dependencies with a Harvard PhD researcher; **peer-reviewed**.

IEEE x CMU / *Electrical Lead, animatronic control system* Mar 2025 – Present

- Led a **6-person** team; architected the **RTOS task system in embedded C** coordinating motors, audio, and sensors.

TECHNICAL SKILLS

AI Systems & Inference Optimization: AI inference, INT8 quantization, pruning, TinyML, edge inference, TensorRT, kernel-level optimization, latency & power profiling, benchmarking, PyTorch, YOLOv8, physics-informed ML, sim-to-real

Signal Processing & Numerical Methods: Fourier & Laplace transforms, DFT/FFT, sampling, filter design, spectral analysis, sensor fusion, Kalman filtering, state estimation, robust regression, time-series analysis, MATLAB

Computer Architecture & Digital Design: SystemVerilog, RTL design, pipelining, hazard detection & forwarding, testbench verification, RISC-V, FPGA, Vivado, QuestaSim

Low-Level & Systems Programming: C/C++, Python, Rust, embedded C firmware, QNX RTOS, Linux, device drivers, I2C/SPI/UART, multithreading, POSIX sockets, CMake, Git, Docker, NumPy/SciPy, OpenCV, ROS 2